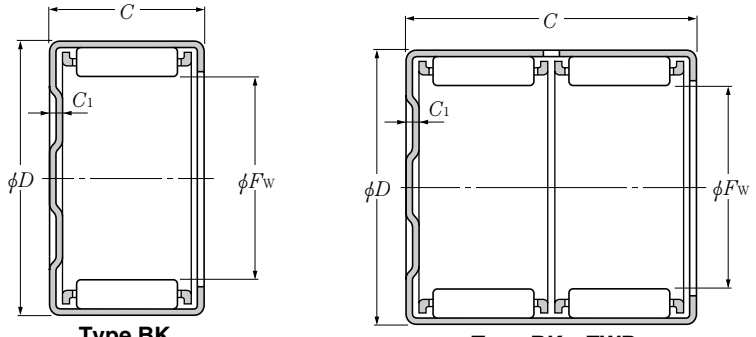


Type BK

Type BK··ZWD

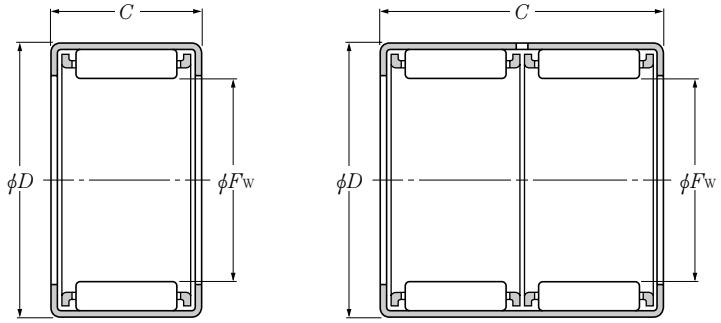
Closed End Type

F_w 20～25mm

Boundary dimensions mm				Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	oil	Bearing numbers open end design	closed end design	Mass kg (approx.)	Appropriate[1] inner ring (as a reference)
F_w	D	C $^{0}_{-0.2}$	C_1 max	C_r	C_{or}	C_r	C_{or}						
	26	30	—	22 200	40 000	2 270	4 100	8 000	12 000	**HK2030ZWD**	—	0.035	IR17×20×30.5
	26	30	2.7	22 200	40 000	2 270	4 100	8 000	12 000	—	**BK2030ZWD**	0.037	IR17×20×30.5
20	27	15	—	13 000	17 300	1 330	1 760	8 000	12 000	**HMK2015**	—	0.021	IR17×20×16.5
	27	20	—	17 700	25 600	1 800	2 610	8 000	12 000	**HMK2020**	—	0.027	IR17×20×20.5
	27	25	—	22 000	34 000	2 240	3 450	8 000	12 000	**HMK2025**	—	0.034	IR15×20×26
	27	30	—	26 100	42 000	2 660	4 300	8 000	12 000	**HMK2030**	—	0.041	IR17×20×30.5
21	29	16	—	15 300	19 100	1 560	1 940	7 500	11 000	**HMK2116**	—	0.027	IR17×21×20
	29	20	—	19 400	25 800	1 970	2 630	7 500	11 000	**HMK2120**	—	0.033	—
	28	12	—	9 750	14 300	995	1 460	7 500	11 000	**HK2212**	—	0.013	IR17×22×13
	28	12	2.7	9 750	14 300	995	1 460	7 500	11 000	—	**BK2212**	0.015	IR17×22×13
	28	16	—	13 600	22 100	1 390	2 250	7 500	11 000	**HK2216C**	—	0.021	IR17×22×18
	28	16	2.7	13 600	22 100	1 390	2 250	7 500	11 000	—	**BK2216**	0.024	IR17×22×18
	28	20	—	17 200	29 800	1 760	3 050	7 500	11 000	**HK2220C**	—	0.026	IR17×22×20.5
22	28	20	2.7	17 200	29 800	1 760	3 050	7 500	11 000	—	**BK2220**	0.030	IR17×22×20.5
	29	10	—	8 400	10 100	855	1 030	7 500	11 000	**HMK2210**	—	0.015	IR17×22×13
	29	15	—	13 400	18 500	1 370	1 890	7 500	11 000	**HMK2215**	—	0.022	IR17×22×16D
	29	20	—	18 200	27 400	1 860	2 790	7 500	11 000	**HMK2220**	—	0.030	IR17×22×20.5
	29	25	—	23 600	38 500	2 410	3 900	7 500	11 000	**HMK2225**	—	0.037	IR17×22×26
	29	30	—	26 900	45 000	2 740	4 600	7 500	11 000	**HMK2230**	—	0.045	IR17×22×32
24	31	20	—	18 800	29 100	1 910	2 970	6 500	10 000	**HMK2420**	—	0.032	—
	31	28	—	26 000	44 500	2 650	4 500	6 500	10 000	**HMK2428**	—	0.045	IR20×24×28.5
	32	12	—	11 800	16 300	1 200	1 660	6 500	9 500	**HK2512**	—	0.021	IR20×25×12.5
	32	12	2.7	11 800	16 300	1 200	1 660	6 500	9 500	—	**BK2512**	0.023	IR20×25×12.5
	32	16	—	15 900	24 000	1 620	2 450	6 500	9 500	**HK2516**	—	0.027	IR20×25×17
	32	16	2.7	15 900	24 000	1 620	2 450	6 500	9 500	—	**BK2516**	0.031	IR20×25×17
25	32	20	—	20 300	33 000	2 070	3 350	6 500	9 500	**HK2520**	—	0.034	IR20×25×20.5
	32	20	2.7	20 300	33 000	2 070	3 350	6 500	9 500	—	**BK2520**	0.039	IR20×25×20.5
	32	26	—	26 400	46 000	2 690	4 700	6 500	9 500	**HK2526**	—	0.045	IR20×25×26.5
	32	26	2.7	26 400	46 000	2 690	4 700	6 500	9 500	—	**BK2526**	0.049	IR20×25×26.5
	32	38	—	35 000	65 500	3 550	6 700	6 500	9 500	**HK2538ZWD**	—	0.065	IR20×25×38.5

Note 1) Bearing with inner ring is represented by HK+IR. (Refer to "Inner Ring Dimensions Table" on page B-119, B120.)
EX. HK2512 + IR20×25×12.5

Type HK, Type HK··ZWD
Type HMK, Type HMK··ZWD
Type BK, Type BK··ZWD



Type HK, Type HMK

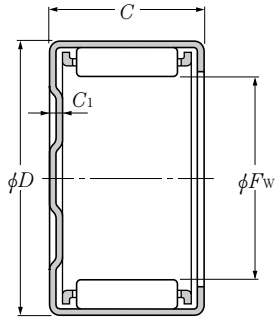
Type HK··ZWD
Type HMK··ZWD

Opened End Type

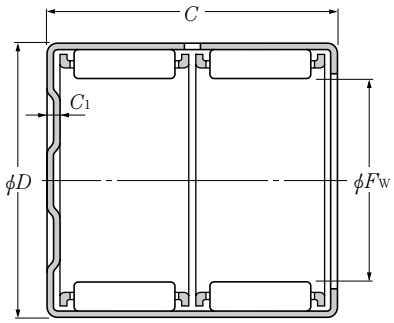
F_w 25～30mm

Boundary dimensions mm				Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	oil	Bearing numbers open end design	closed end design	Mass kg (approx.)	Appropriate[1] inner ring (as a reference)
F_w	D	C $^{0}_{-0.2}$	C_1 max	C_r	C_{or}	C_r	C_{or}						
25	32	38	2.7	35 000	65 500	3 550	6 700	6 500	9 500	—	**BK2538ZWD**	0.069	IR20×25×38.5
	33	10	—	9 150	10 400	935	1 060	6 500	9 500	**HMK2510**	—	0.019	IR20×25×12.5
	33	15	—	15 200	19 900	1 550	2 030	6 500	9 500	**HMK2515C**	—	0.029	IR20×25×16
	33	20	—	21 800	31 500	2 220	3 200	6 500	9 500	**HMK2520**	—	0.039	IR20×25×20.5
	33	25	—	26 700	41 000	2 720	4 200	6 500	9 500	**HMK2525**	—	0.048	IR20×25×26.5
	33	30	—	32 500	53 000	3 300	5 400	6 500	9 500	**HMK2530**	—	0.058	IR20×25×32
26	34	16	—	17 200	23 300	1 750	2 380	6 000	9 000	**HMK2616**	—	0.032	IR22×26×20
	34	20	—	21 800	31 500	2 220	3 200	6 000	9 000	**HMK2620**	—	0.040	—
28	35	16	—	17 300	27 600	1 760	2 820	5 500	8 500	**HK2816**	—	0.030	IR22×28×17
	35	16	2.7	17 300	27 600	1 760	2 820	5 500	8 500	—	**BK2816**	0.034	IR22×28×17
	35	20	—	21 300	36 000	2 170	3 700	5 500	8 500	**HK2820**	—	0.038	IR22×28×20.5
	35	20	2.7	21 300	36 000	2 170	3 700	5 500	8 500	—	**BK2820**	0.043	IR22×28×20.5
	37	20	—	23 600	32 500	2 410	3 350	5 500	8 500	**HMK2820**	—	0.049	IR22×28×20.5
	37	30	—	35 000	54 500	3 600	5 550	5 500	8 500	**HMK2830**	—	0.073	—
29	38	20	—	24 600	35 000	2 510	3 550	5 500	8 500	**HMK2920**	—	0.050	—
	38	30	—	34 500	54 000	3 550	5 550	5 500	8 500	**HMK2930**	—	0.075	—
30	37	12	—	13 000	19 500	1 320	1 990	5 500	8 000	**HK3012**	—	0.024	IR25×30×12.5
	37	12	2.7	13 000	19 500	1 320	1 990	5 500	8 000	—	**BK3012**	0.028	IR25×30×12.5
	37	16	—	18 100	30 000	1 850	3 050	5 500	8 000	**HK3016**	—	0.032	IR25×30×17
	37	16	2.7	18 100	30 000	1 850	3 050	5 500	8 000	—	**BK3016C**	0.037	IR25×30×17
	37	20	—	22 300	39 500	2 280	4 000	5 500	8 000	**HK3020**	—	0.040	IR25×30×20.5
	37	20	2.7	22 300	39 500	2 280	4 000	5 500	8 000	—	**BK3020**	0.047	IR25×30×20.5
	37	26	—	28 500	54 000	2 910	5 500	5 500	8 000	**HK3026**	—	0.053	IR25×30×26.5
	37	26	2.7	29 100	55 000	2 960	5 650	5 500	8 000	—	**BK3026**	0.059	IR25×30×26.5
	37	38	—	38 500	78 500	3 900	8 000	5 500	8 000	**HK3038ZWD**	—	0.076	IR25×30×38.5
	37	38	2.7	38 500	78 500	3 900	8 000	5 500	8 000	—	**BK3038ZWD**	0.083	IR25×30×38.5
	40	13	—	14 100	17 100	1 430	1 750	5 500	8 000	**HMK3013**	—	0.040	IR25×30×16
	40	15	—	17 100	22 100	1 750	2 250	5 500	8 000	**HMK3015**	—	0.044	IR25×30×16
	40	20	—	24 200	34 500	2 470	3 500	5 500	8 000	**HMK3020**	—	0.058	IR25×30×20.5
	40	25	—	31 000	47 000	3 150	4 800	5 500	8 000	**HMK3025**	—	0.073	IR25×30×26.5

Note 1) Bearing with inner ring is represented by HK+IR. (Refer to "Inner Ring Dimensions Table" on page B-120.)
EX. HK2820 + IR22×28×20.5



Type BK



Type BK··ZWD

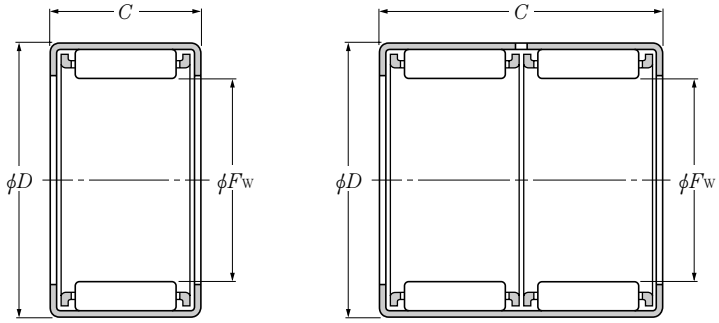
Closed End Type

F_w 30～40mm

Boundary dimensions mm				Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	oil	Bearing numbers open end design	closed end design	Mass kg (approx.)	Appropriate[1] inner ring (as a reference)
F_w	D	C $^{0}_{-0.2}$	C_1 max	C_r	C_{or}	C_r	C_{or}						
30	40	30	—	36 000	57 500	3 700	5 850	5 500	8 000	**HMK3030**	—	0.087	IR25×30×32
32	42	20	—	27 500	38 000	2 800	3 850	5 000	7 500	**HMK3220**	—	0.062	—
	42	30	—	41 500	64 500	4 250	6 550	5 000	7 500	**HMK3230**	—	0.092	—
35	42	12	—	14 000	22 800	1 430	2 320	4 700	7 000	**HK3512**	—	0.028	—
	42	12	2.7	14 000	22 800	1 430	2 320	4 700	7 000	—	**BK3512**	0.033	—
	42	16	—	19 700	35 000	2 000	3 600	4 700	7 000	**HK3516**	—	0.037	—
	42	16	2.7	19 700	35 000	2 000	3 600	4 700	7 000	—	**BK3516**	0.044	—
	42	20	—	24 800	47 500	2 530	4 850	4 700	7 000	**HK3520**	—	0.046	—
	42	20	2.7	24 800	47 500	2 530	4 850	4 700	7 000	—	**BK3520**	0.055	—
	45	12	—	14 900	17 600	1 520	1 800	4 700	7 000	**HMK3512**	—	0.040	—
	45	15	—	20 200	26 200	2 060	2 670	4 700	7 000	**HMK3515**	—	0.050	—
	45	20	—	28 400	40 500	2 890	4 100	4 700	7 000	**HMK3520**	—	0.067	—
	45	25	—	36 000	54 500	3 650	5 550	4 700	7 000	**HMK3525**	—	0.083	—
	45	30	—	43 000	69 000	4 400	7 000	4 700	7 000	**HMK3530**	—	0.100	—
37	47	20	—	29 300	43 000	2 990	4 350	4 300	6 500	**HMK3720**	—	0.070	—
	47	30	—	44 500	73 000	4 550	7 450	4 300	6 500	**HMK3730**	—	0.105	—
38	48	15	—	21 700	29 300	2 210	2 990	4 300	6 500	**HMK3815**	—	0.054	—
	48	20	—	30 500	45 000	3 100	4 600	4 300	6 500	**HMK3820**	—	0.072	—
	48	25	—	38 500	61 000	3 900	6 250	4 300	6 500	**HMK3825**	—	0.090	—
	48	30	—	46 000	77 000	4 700	7 850	4 300	6 500	**HMK3830**	—	0.107	IR32×38×32
	48	45	—	62 000	113 000	6 300	11 500	4 300	6 500	**HMK3845ZWD**	—	0.161	—
40	47	12	—	15 100	26 000	1 540	2 660	4 000	6 000	**HK4012**	—	0.031	IR35×40×12.5
	47	12	2.7	15 100	26 000	1 540	2 660	4 000	6 000	—	**BK4012**	0.038	IR35×40×12.5
	47	16	—	21 100	40 000	2 150	4 100	4 000	6 000	**HK4016**	—	0.041	IR35×40×17
	47	16	2.7	21 100	40 000	2 150	4 100	4 000	6 000	—	**BK4016**	0.051	IR35×40×17
	47	20	—	25 900	52 500	2 650	5 350	4 000	6 000	**HK4020**	—	0.052	IR35×40×20.5
	47	20	2.7	25 900	52 500	2 650	5 350	4 000	6 000	—	**BK4020**	0.064	IR35×40×20.5
	50	15	—	23 100	32 500	2 350	3 300	4 000	6 000	**HMK4015**	—	0.056	IR35×40×17
	50	20	—	32 500	50 000	3 300	5 100	4 000	6 000	**HMK4020**	—	0.075	IR35×40×20.5
	50	25	—	41 000	67 500	4 150	6 900	4 000	6 000	**HMK4025**	—	0.094	—

Note 1) Bearing with inner ring is represented by HK+IR. (Refer to "Inner Ring Dimensions Table" on page B-121, B122.)
EX. HK4012 + IR35×40×12.5

Type HK
Type HMK, Type HMK··ZWD
Type BK



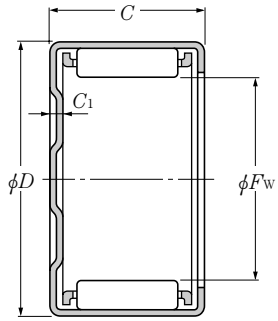
Type HK, Type HMK　　Type HMK··ZWD

Opened End Type

F_w 40～50mm

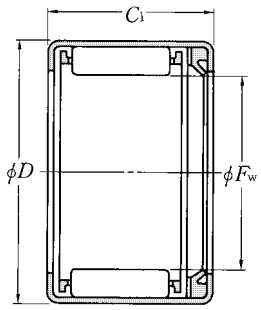
Boundary dimensions mm				Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	oil	Bearing numbers open end design	Bearing numbers closed end design	Mass kg (approx.)	Appropriate[1] inner ring (as a reference)
F_w	D	C $^{0}_{-0.2}$	C_1 max	C_r	C_{or}	C_r	C_{or}						
40	50	30	—	49 000	85 000	5 000	8 700	4 000	6 000	**HMK4030**	—	0.112	IR35×40×34
	50	40	—	58 500	107 000	5 950	10 900	4 000	6 000	**HMK4040ZWD**	—	0.150	—
45	52	16	—	21 600	43 000	2 210	4 400	3 700	5 500	**HK4516**	—	0.046	IR40×45×17
	52	16	2.7	21 600	43 000	2 210	4 400	3 700	5 500	—	**BK4516**	0.058	IR40×45×17
	52	20	—	27 600	59 000	2 810	6 000	3 700	5 500	**HK4520**	—	0.058	IR40×45×20.5
	52	20	2.7	27 600	59 000	2 810	6 000	3 700	5 500	—	**BK4520**	0.072	IR40×45×20.5
	55	20	—	33 000	53 000	3 350	5 400	3 700	5 500	**HMK4520**	—	0.083	IR40×45×20.5
	55	25	—	41 500	71 500	4 250	7 300	3 700	5 500	**HMK4525**	—	0.104	IR40×45×26.5
	55	30	—	49 500	90 000	5 050	9 150	3 700	5 500	**HMK4530**	—	0.125	IR40×45×34
	55	40	—	59 500	113 000	6 050	11 500	3 700	5 500	**HMK4540ZWD**	—	0.167	—
50	58	20	—	31 500	63 000	3 200	6 450	3 200	4 800	**HK5020**	—	0.072	IR40×50×22
	58	20	2.7	31 500	63 000	3 200	6 450	3 200	4 800	—	**BK5020**	0.087	IR40×50×22
	58	25	—	38 500	82 000	3 900	8 400	3 200	4 800	**HK5025**	—	0.090	IR45×50×25.5
	58	25	2.7	38 500	82 000	3 900	8 400	3 200	4 800	—	**BK5025**	0.109	IR45×50×25.5
	62	12	—	18 200	23 600	1 860	2 410	3 200	4 800	**HMK5012**	—	0.067	—
	62	15	—	25 900	37 000	2 650	3 800	3 200	4 800	**HMK5015**	—	0.084	—
	62	20	—	37 500	60 000	3 850	6 100	3 200	4 800	**HMK5020**	—	0.112	IR40×50×22
	62	25	—	48 000	82 500	4 900	8 450	3 200	4 800	**HMK5025**	—	0.140	IR45×50×25.5
	62	30	—	58 500	105 000	5 950	10 700	3 200	4 800	**HMK5030B**	—	0.168	IR45×50×32
	62	40	—	70 000	134 000	7 150	13 600	3 200	4 800	**HMK5040ZWD**	—	0.224	—
	62	45	—	79 000	156 000	8 050	15 900	3 200	4 800	**HMK5045ZWBD**	—	0.252	—

Note 1) Bearing with inner ring is represented by HK+IR. (Refer to "Inner Ring Dimensions Table" on page B-122, B123.)
EX. HK4516 + IR40×45×17

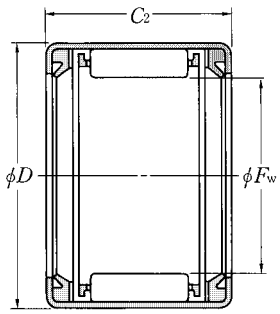


Type BK
Closed End Type

Type HK··L
Type HMK··L
Type HK··LL
Type HMK··LL
Type BK··L



Type HK··L, Type HMK··L
(Opened end and single-side seal type)

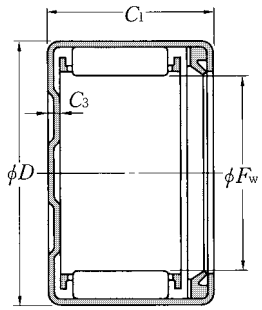


Type HK··LL, Type HMK··LL
(Opened end and double-side seal type)

F_w 8～25mm

Boundary dimensions mm					Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	Bearing numbers		
F_w	D	C_1 $^{0}_{-0.2}$	C_2 $^{0}_{-0.2}$	C_3 max	C_r	C_{or}	C_r	C_{or}		open end single seal	open end double seal	closed end single seal
8	15	12	14	—	3 800	2 870	390	293	10 000	**HMK0812L**	**HMK0814LL**	—
10	17	12	14	—	4 250	3 450	435	350	10 000	**HMK1012L**	**HMK1014LL**	—
12	18	14	16	2.7	6 600	7 300	675	745	10 000	**HK 1214L**	**HK 1216LL**	**BK1214L**
	19	14	16	—	7 100	6 900	725	705	10 000	**HMK1214L**	**HMK1216LL**	—
14	20	14	16	2.7	7 200	8 500	735	865	10 000	**HK 1414L**	**HK 1416LL**	**BK1414L**
	22	19	22	—	11 500	12 000	1 180	1 220	10 000	**HMK1419L**	**HMK1422LL**	—
15	21	14	16	2.7	7 500	9 100	765	930	10 000	**HK 1514L**	**HK 1516LL**	**BK1514L**
	22	13	16	—	6 100	6 000	620	610	10 000	**HMK1513L**	**HMK1516LL**	—
	22	18	21	—	10 900	12 700	1 120	1 300	10 000	**HMK1518L**	**HMK1521LL**	—
16	22	14	16	2.7	7 750	9 700	795	990	10 000	**HK 1614L**	**HK 1616LL**	**BK1614L**
	24	23	26	—	15 600	18 200	1 590	1 860	10 000	**HMK1623L**	**HMK1626LL**	—
17	24	18	21	—	12 100	15 000	1 230	1 530	9 500	**HMK1718L**	**HMK1721LL**	—
18	24	14	16	2.7	8 300	10 900	845	1 110	9 000	**HK 1814L**	**HK 1816LL**	**BK1814L**
	25	18	21	—	12 000	15 100	1 220	1 540	9 000	**HMK1818L**	**HMK1821LL**	—
	25	20	23	—	13 800	18 000	1 400	1 830	9 000	**HMK1820L**	**HMK1823LL**	—
19	27	19	22	—	13 900	16 300	1 410	1 660	8 500	**HMK1919L**	**HMK1922LL**	—
20	26	—	16	—	9 250	13 000	945	1 330	8 000	—	**HK 2016LL**	—
	26	18	20	2.7	13 000	20 100	1 320	2 050	8 000	**HK 2018L**	**HK 2020LL**	**BK2018L**
	27	18	21	—	13 000	17 300	1 330	1 760	8 000	**HMK2018L**	**HMK2021LL**	—
	27	23	26	—	17 700	25 600	1 800	2 610	8 000	**HMK2023L**	**HMK2026LL**	—
22	28	—	16	—	9 750	14 300	995	1 460	7 500	—	**HK 2216LL**	—
	28	18	20	2.7	13 600	22 100	1 390	2 250	7 500	**HK 2218L**	**HK 2220LL**	**BK2218L**
	29	18	21	—	13 400	18 500	1 370	1 890	7 500	**HMK2218L**	**HMK2221LL**	—
	29	23	26	—	18 200	27 400	1 860	2 790	7 500	**HMK2223L**	**HMK2226LL**	—
24	31	23	26	—	18 800	29 100	1 910	2 970	6 500	**HMK2423L**	**HMK2426LL**	—
25	32	—	16	—	11 800	16 300	1 200	1 660	6 500	—	**HK 2516LL**	—
	32	18	20	2.7	15 900	24 000	1 620	2 450	6 500	**HK 2518L**	**HK 2520LL**	**BK2518L**

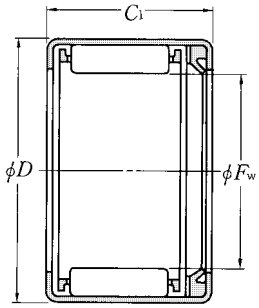
Note 1) Bearing with inner ring is represented by HK+IR. (Refer to "Inner Ring Dimensions Table" on page B-117 to B119.)
EX. HK1416LL + IR10×14×20



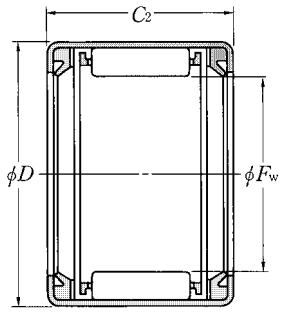
Type BK··L
(Closed end and single-side seal type)

Mass kg (approx.)			Appropriate[1] inner ring (as a reference)	
open end single seal	open end double seal	closed end single seal	single seal	double seal
0.0071	0.0075	—	IR 5×8×16	IR 5× 8×16
0.0084	0.0089	—	IR 7×10×16	IR 7×10×16
0.011	0.0120	0.012	IR 9×12×16	—
0.011	0.0120	—	IR 9×12×16	—
0.012	0.0140	0.014	IR10×14×16	IR10×14×20
0.020	0.0210	—	IR10×14×20	—
0.013	0.0140	0.014	IR12×15×16.5	IR12×15×16.5
0.014	0.0150	—	IR12×15×16	IR12×15×16.5
0.017	0.0180	—	IR12×15×22.5	IR12×15×22.5
0.013	0.0150	0.015	IR12×16×16	IR12×16×20
0.028	0.0290	—	—	—
0.019	0.0200	—	IR12×17×20.5	IR12×17×25.5
0.015	0.0170	0.017	IR15×18×16	IR15×18×17.5
0.020	0.0210	—	IR15×18×20.5	IR15×18×25.5
0.023	0.0240	—	IR15×18×20.5	IR15×18×25.5
0.027	0.0290	—	IR15×19×20	—
—	0.0190	—	—	IR15×20×18
0.021	0.0240	0.024	IR17×20×20	IR17×20×20.5
0.022	0.0240	—	IR17×20×20	IR15×20×23
0.029	0.0310	—	IR15×20×26	IR17×20×30.5
—	0.0200	—	—	IR17×22×18
0.024	0.0260	0.027	IR17×22×20.5	IR17×22×23
0.024	0.0260	—	IR17×22×20.5	IR17×22×23
0.032	0.0330	—	IR17×22×26	—
0.035	0.0370	—	—	IR20×24×28.5
—	0.0270	—	—	IR20×25×18D
0.031	0.0330	0.035	IR20×25×20	IR20×25×23

Type HK··L
Type HMK··L
Type HK··LL
Type HMK··LL
Type BK··L



Type HK··L, Type HMK··L
(Opened end and single-side seal type)

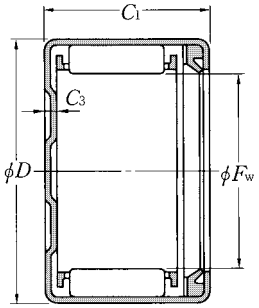


Type HK··LL, Type HMK··LL
(Opened end and double-side seal type)

F_w 25～50mm

Boundary dimensions mm					Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	Bearing numbers open end single seal	open end double seal	closed end single seal
F_w	D	C_1 $^{0}_{-0.2}$	C_2 $^{0}_{-0.2}$	C_3 max	C_r	C_{or}	C_r	C_{or}				
25	33	18	21	—	15 800	21 000	1 610	2 140	6 500	**HMK2518L**	**HMK2521LL**	— —
	33	23	26	—	21 800	31 500	2 220	3 200	6 500	**HMK2523L**	**HMK2526LL**	— —
28	35	—	20	—	17 300	27 600	1 760	2 820	5 500	— —	**HK 2820LL**	— —
	37	23	26	—	23 600	32 500	2 410	3 350	5 500	**HMK2823L**	**HMK2826LL**	— —
30	37	—	16	—	13 000	19 500	1 320	1 990	5 500	— —	**HK 3016LL**	— —
	37	18	20	2.7	18 100	30 000	1 850	3 050	5 500	**HK 3018L**	**HK 3020LL**	**BK3018L**
	40	23	26	—	24 200	34 500	2 470	3 500	5 500	**HMK3023L**	**HMK3026LL**	— —
	40	28	31	—	31 000	47 000	3 150	4 800	5 500	**HMK3028L**	**HMK3031LL**	— —
32	42	23	26	—	27 500	38 000	2 800	3 850	5 000	**HMK3223L**	**HMK3226LL**	— —
35	42	—	16	—	14 000	22 800	1 430	2 320	4 600	— —	**HK 3516LL**	— —
	42	18	20	2.7	19 700	35 000	2 000	3 600	4 600	**HK 3518L**	**HK 3520LL**	**BK3518L**
	45	18	21	—	20 200	26 200	2 060	2 670	4 600	**HMK3518L**	**HMK3521LL**	— —
	45	28	31	—	36 000	54 500	3 650	5 550	4 600	**HMK3528L**	**HMK3531LL**	— —
38	48	28	31	—	38 500	61 000	3 900	6 250	4 200	**HMK3828L**	**HMK3831LL**	— —
40	47	—	16	—	15 100	26 000	1 540	2 660	4 000	— —	**HK 4016LL**	— —
	47	18	20	2.7	21 100	40 000	2 150	4 100	4 000	**HK 4018L**	**HK 4020LL**	**BK4018L**
	50	18	21	—	23 100	32 500	2 350	3 300	4 000	**HMK4018L**	**HMK4021LL**	— —
	50	28	31	—	41 000	67 500	4 150	6 900	4 000	**HMK4028L**	**HMK4031LL**	— —
45	52	18	20	2.7	21 600	43 000	2 210	4 400	3 600	**HK 4518L**	**HK 4520LL**	**BK4518L**
	55	23	26	—	33 000	53 000	3 350	5 400	3 600	**HMK4523L**	**HMK4526LL**	— —
50	58	22	24	2.7	31 500	63 000	3 200	6 450	3 200	**HK 5022L**	**HK 5024LL**	**BK5022L**
	62	28	31	—	48 000	82 500	4 900	8 450	3 200	**HMK5028L**	**HMK5031LL**	— —

Note 1) Bearing with inner ring is represented by HK+IR. (Refer to "Inner Ring Dimensions Table" on page B-119 to B123.)
EX. HK5022L + IR45×50×25

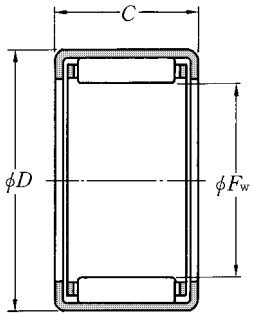


Type BK··L
(Closed end and single-side seal type)

Mass kg (approx.)			Appropriate[1] inner ring (as a reference)	
open end single seal	open end double seal	closed end single seal	single seal	double seal
0.031	0.034	—	IR20×25×20	IR20×25×23
0.041	0.043	—	IR20×25×26.5	IR20×25×26.5
—	0.037	—	—	IR22×28×23
0.051	0.054	—	—	IR22×28×30
—	0.027	—	—	IR25×30×18D
0.037	0.039	0.045	IR25×30×20	IR25×30×23
0.061	0.064	—	IR25×30×26	IR25×30×26.5
0.076	0.078	—	IR25×30×30	IR25×30×32
0.065	0.069	—	—	IR28×32×30
—	0.036	—	—	—
0.037	0.040	0.047	—	—
0.053	0.056	—	—	—
0.086	0.089	—	—	—
0.094	0.098	—	IR32×38×32	IR32×38×32
—	0.041	—	—	IR35×40×17
0.047	0.050	0.062	IR35×40×20	—
0.060	0.063	—	IR35×40×20	—
0.097	0.100	—	IR35×40×30	IR32×40×36
0.054	0.057	0.072	IR40×45×20	—
0.087	0.091	—	IR40×45×26.5	IR40×45×26.5
0.086	0.089	0.104	IR45×50×25	IR45×50×25.5
0.144	0.149	—	IR45×50×32	IR45×50×32

Inch series

Type DCL



F_w 6.350～15.875mm

Boundary dimensions mm ($\frac{1}{25.4}$mm)			Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	Limiting speeds r/min oil	Bearing numbers	Mass kg	Appropriate[1] inner ring
F_w	D	C $^{0}_{-0.2}$	C_r	C_{or}	C_r	C_{or}				(approx.)	(as a reference)
	11.112(7/16)	6.350(1/4)	1 580	1 110	161	113	25 000	38 000	DCL 44T2	0.0022	—
6.350(1/4)	11.112(7/16)	7.938(5/16)	2 160	1 670	221	170	25 000	38 000	DCL 45T2	0.0033	—
	11.112(7/16)	11.112(7/16)	3 550	3 150	360	320	25 000	38 000	DCL 47T2	0.0038	—
	12.700(1/2)	7.938(5/16)	2 940	2 610	300	266	20 000	30 000	DCL 55	0.0032	—
7.938(5/16)	12.700(1/2)	9.525(3/8)	3 900	3 750	400	385	20 000	30 000	DCL 56	0.0039	—
	12.700(1/2)	11.112(7/16)	4 800	4 950	490	505	20 000	30 000	DCL 57	0.0048	—
	12.700(1/2)	14.288(9/16)	6 500	7 250	665	740	20 000	30 000	DCL 59	0.0058	—
	14.288(9/16)	7.938(5/16)	3 100	2 910	315	297	17 000	25 000	DCL 65	0.0037	—
9.525(3/8)	14.288(9/16)	9.525(3/8)	4 100	4 200	420	430	17 000	25 000	DCL 66	0.0045	—
	14.288(9/16)	12.700(1/2)	5 900	6 650	600	675	17 000	25 000	DCL 68	0.0065	—
	14.288(9/16)	15.875(5/8)	7 500	9 050	765	925	17 000	25 000	DCL 610	0.0075	—
11.112(7/16)	15.875(5/8)	12.700(1/2)	6 450	7 800	660	795	15 000	22 000	DCL 78	0.0068	—
	17.462(11/16)	7.938(5/16)	3 550	3 700	360	380	13 000	19 000	DCL 85	0.0047	—
	17.462(11/16)	9.525(3/8)	4 700	5 350	480	550	13 000	19 000	DCL 86	0.0057	—
12.700(1/2)	17.462(11/16)	11.112(7/16)	5 800	7 050	590	715	13 000	19 000	DCL 87	0.0066	—
	17.462(11/16)	12.700(1/2)	6 700	8 500	685	865	13 000	19 000	DCL 88	0.0080	—
	17.462(11/16)	15.875(5/8)	8 550	11 600	870	1 180	13 000	19 000	DCL 810	0.0095	—
	17.462(11/16)	19.050(3/4)	10 400	14 900	1 060	1 520	13 000	19 000	DCL 812	0.0120	—
	19.050(3/4)	7.938(5/16)	3 800	4 250	390	430	11 000	17 000	DCL 95	0.0052	—
	19.050(3/4)	9.525(3/8)	5 050	6 150	515	625	11 000	17 000	DCL 96	0.0063	MI-060908
14.288(9/16)	19.050(3/4)	11.112(7/16)	6 250	8 000	635	815	11 000	17 000	DCL 97	0.0073	MI-060908
	19.050(3/4)	12.700(1/2)	7 200	9 650	735	985	11 000	17 000	DCL 98	0.0086	MI-060908
	19.050(3/4)	15.875(5/8)	9 200	13 200	935	1 350	11 000	17 000	DCL 910	0.0110	—
	19.050(3/4)	19.050(3/4)	11 200	17 000	1 140	1 730	11 000	17 000	DCL 912	0.0130	—
	20.638(13/16)	7.938(5/16)	4 050	4 750	415	485	10 000	15 000	DCL 105	0.0075	—
	20.638(13/16)	11.112(7/16)	6 650	9 000	680	915	10 000	15 000	DCL 107	0.0080	—
15.875(5/8)	20.638(13/16)	12.700(1/2)	7 700	10 800	785	1 110	10 000	15 000	DCL 108	0.0091	—
	20.638(13/16)	15.875(5/8)	9 800	14 800	1 000	1 510	10 000	15 000	DCL1010	0.0130	MI-061012
	20.638(13/16)	19.050(3/4)	11 900	19 000	1 220	1 940	10 000	15 000	DCL1012	0.0140	MI-061012

Note 1) Bearing with inner ring is represented by DCL-MI. (Refer to Inner Ring Dimension Table on page B-129.)
EX. DCL96 + MI-060908
Remarks: Manufacture of the closed end type bearings under this Table is also available.

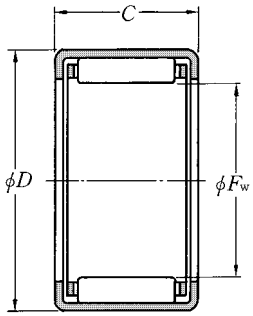
F_w 15.875～25.400mm

Boundary dimensions mm ($\frac{1}{25.4}$mm) F_w	D	C $^{0}_{-0.2}$	Basic load ratings dynamic N C_r	static N C_{or}	dynamic kgf C_r	static kgf C_{or}	Limiting speeds r/min grease	oil	Bearing numbers	Mass kg (approx.)	Appropriate[1] inner ring (as a reference)
15.875($\frac{5}{8}$)	20.638($\frac{13}{16}$)	22.225($\frac{7}{8}$)	14 100	23 500	1 430	2 400	10 000	15 000	**DCL1014**	0.0160	MI-061016
17.462($\frac{11}{16}$)	22.225($\frac{7}{8}$)	9.525($\frac{3}{8}$)	5 700	7 650	585	780	9 500	14 000	**DCL 116**	0.0075	—
	22.225($\frac{7}{8}$)	12.700($\frac{1}{2}$)	8 150	12 000	830	1 230	9 500	14 000	**DCL 118**	0.0110	—
	22.225($\frac{7}{8}$)	15.875($\frac{5}{8}$)	10 400	16 400	1 060	1 680	9 500	14 000	**DCL1110**	0.0130	—
	22.225($\frac{7}{8}$)	19.050($\frac{3}{4}$)	12 600	21 100	1 290	2 150	9 500	14 000	**DCL1112**	0.0160	—
19.050($\frac{3}{4}$)	25.400(1)	9.525($\frac{3}{8}$)	6 450	6 950	660	705	8 500	13 000	**DCL 126**	0.0110	—
	25.400(1)	12.700($\frac{1}{2}$)	9 800	11 900	1 000	1 210	8 500	13 000	**DCL 128**	0.0140	MI-081210
	25.400(1)	15.875($\frac{5}{8}$)	12 900	16 800	1 310	1 720	8 500	13 000	**DCL1210**	0.0170	MI-081210
	25.400(1)	19.050($\frac{3}{4}$)	15 900	22 100	1 620	2 250	8 500	13 000	**DCL1212**	0.0210	MI-081212
	25.400(1)	22.225($\frac{7}{8}$)	19 000	27 700	1 930	2 830	8 500	13 000	**DCL1214**	0.0260	MI-081216
	25.400(1)	25.400(1)	21 700	33 000	2 210	3 350	8 500	13 000	**DCL1216**	0.0300	MI-081216
20.638($\frac{13}{16}$)	26.988(1 $\frac{1}{16}$)	9.525($\frac{3}{8}$)	6 950	7 800	710	795	8 000	12 000	**DCL 136**	0.0120	—
	26.988(1 $\frac{1}{16}$)	12.700($\frac{1}{2}$)	10 600	13 400	1 080	1 370	8 000	12 000	**DCL 138**	0.0160	—
	26.988(1 $\frac{1}{16}$)	15.875($\frac{5}{8}$)	13 900	19 000	1 410	1 930	8 000	12 000	**DCL1310**	0.0200	—
	26.988(1 $\frac{1}{16}$)	19.050($\frac{3}{4}$)	17 100	24 900	1 750	2 540	8 000	12 000	**DCL1312**	0.0230	—
	26.988(1 $\frac{1}{16}$)	22.225($\frac{7}{8}$)	20 400	31 500	2 080	3 200	8 000	12 000	**DCL1314**	0.0280	—
	26.988(1 $\frac{1}{16}$)	25.400(1)	23 400	37 000	2 380	3 800	8 000	12 000	**DCL1316**	0.0320	—
	26.988(1 $\frac{1}{16}$)	31.750(1 $\frac{1}{4}$)	29 000	49 000	2 960	5 000	8 000	12 000	**DCL1320**	0.0400	—
22.225($\frac{7}{8}$)	28.575(1 $\frac{1}{8}$)	9.525($\frac{3}{8}$)	7 150	8 300	730	845	7 500	11 000	**DCL 146**	0.0130	MI-101406
	28.575(1 $\frac{1}{8}$)	12.700($\frac{1}{2}$)	10 900	14 200	1 110	1 450	7 500	11 000	**DCL 148**	0.0170	MI-101408
	28.575(1 $\frac{1}{8}$)	19.050($\frac{3}{4}$)	17 600	26 400	1 800	2 700	7 500	11 000	**DCL1412**	0.0250	MI-101412
	28.575(1 $\frac{1}{8}$)	22.225($\frac{7}{8}$)	21 000	33 000	2 140	3 400	7 500	11 000	**DCL1414**	0.0340	MI-101416
	28.575(1 $\frac{1}{8}$)	25.400(1)	24 100	39 500	2 450	4 000	7 500	11 000	**DCL1416**	0.0340	MI-101416
23.812($\frac{15}{16}$)	30.162(1 $\frac{3}{16}$)	15.875($\frac{5}{8}$)	14 600	21 300	1 490	2 170	6 500	10 000	**DCL1510**	0.0230	—
	30.162(1 $\frac{3}{16}$)	25.400(1)	24 700	41 500	2 520	4 250	6 500	10 000	**DCL1516**	0.0360	—
25.400(1)	31.750(1 $\frac{1}{4}$)	9.525($\frac{3}{8}$)	7 550	9 250	770	940	6 500	9 500	**DCL 166**	0.0140	—
	31.750(1 $\frac{1}{4}$)	12.700($\frac{1}{2}$)	11 500	15 800	1 170	1 610	6 500	9 500	**DCL 168**	0.0190	—
	31.750(1 $\frac{1}{4}$)	19.050($\frac{3}{4}$)	18 600	29 500	1 890	3 000	6 500	9 500	**DCL1612**	0.0310	MI-121612
	31.750(1 $\frac{1}{4}$)	22.225($\frac{7}{8}$)	22 100	37 000	2 260	3 750	6 500	9 500	**DCL1614**	0.0340	MI-121616

Note 1) Bearing with inner ring is represented by DCL-MI. (Refer to Inner Ring Dimension Table on page B-129.)
EX. DCL128 + MI-081210
Remarks: Manufacture of the closed end type bearings under this Table is also available.

Inch series

Type DCL



F_w 25.400～41.275mm

Boundary dimensions mm (1/25.4mm) F_w	D	C $^{0}_{-0.2}$	Basic load ratings dynamic N C_r	static N C_{or}	dynamic kgf C_r	static kgf C_{or}	Limiting speeds r/min grease	oil	Bearing numbers	Mass kg (approx.)	Appropriate[1] inner ring (as a reference)
25.400(1)	31.750(1 1/4)	25.400(1)	25 400	44 000	2 590	4 500	6 500	9 500	**DCL1616**	0.038	MI-121616
	31.750(1 1/4)	31.750(1 1/4)	31 500	58 000	3 200	5 900	6 500	9 500	**DCL1620**	0.048	—
26.988(1 1/16)	33.338(1 5/16)	15.875(5/8)	15 900	24 600	1 620	2 510	6 000	9 000	**DCL1710**	0.025	—
28.575(1 1/8)	34.925(1 3/8)	9.525(3/8)	8 150	10 600	830	1 080	5 500	8 500	**DCL 186**	0.016	MI-141808
	34.925(1 3/8)	12.700(1/2)	12 400	18 200	1 260	1 850	5 500	8 500	**DCL 188**	0.021	MI-141808
	34.925(1 3/8)	19.050(3/4)	20 100	34 000	2 050	3 450	5 500	8 500	**DCL1812**	0.032	MI-141812
	34.925(1 3/8)	25.400(1)	27 400	50 500	2 790	5 150	5 500	8 500	**DCL1816**	0.043	MI-141816
	34.925(1 3/8)	31.750(1 1/4)	34 000	66 500	3 450	6 800	5 500	8 500	**DCL1820**	0.053	MI-141820
30.162(1 3/16)	38.100(1 1/2)	25.400(1)	33 000	54 000	3 350	5 500	5 500	8 000	**DCL1916**	0.057	—
31.750(1 1/4)	38.100(1 1/2)	12.700(1/2)	12 500	19 000	1 280	1 940	5 000	7 500	**DCL 208**	0.023	—
	38.100(1 1/2)	15.875(5/8)	16 400	27 000	1 670	2 750	5 000	7 500	**DCL2010**	0.029	—
	38.100(1 1/2)	19.050(3/4)	20 300	35 500	2 070	3 600	5 000	7 500	**DCL2012**	0.036	—
	38.100(1 1/2)	25.400(1)	27 700	53 000	2 830	5 400	5 000	7 500	**DCL2016**	0.047	—
	38.100(1 1/2)	31.750(1 1/4)	34 500	70 000	3 500	7 100	5 000	7 500	**DCL2020**	0.058	—
34.925(1 3/8)	41.275(1 5/8)	12.700(1/2)	13 400	21 400	1 360	2 180	4 700	7 000	**DCL 228**	0.027	—
	41.275(1 5/8)	19.050(3/4)	21 700	40 000	2 210	4 050	4 700	7 000	**DCL2212**	0.038	—
	41.275(1 5/8)	25.400(1)	29 600	59 500	3 000	6 050	4 700	7 000	**DCL2216**	0.051	—
	41.275(1 5/8)	31.750(1 1/4)	36 500	78 500	3 750	8 000	4 700	7 000	**DCL2220**	0.064	—
38.100(1 1/2)	47.625(1 7/8)	12.700(1/2)	17 100	22 800	1 750	2 320	4 300	6 500	**DCL 248**	0.043	—
	47.625(1 7/8)	15.875(5/8)	21 000	29 700	2 150	3 050	4 300	6 500	**DCL2410**	0.054	—
	47.625(1 7/8)	19.050(3/4)	26 600	40 000	2 710	4 100	4 300	6 500	**DCL2412**	0.065	—
	47.625(1 7/8)	22.225(7/8)	32 000	50 500	3 250	5 150	4 300	6 500	**DCL2414**	0.076	MI-202416
	47.625(1 7/8)	25.400(1)	36 500	60 500	3 750	6 200	4 300	6 500	**DCL2416**	0.087	MI-202416
	47.625(1 7/8)	31.750(1 1/4)	46 500	82 000	4 750	8 350	4 300	6 500	**DCL2420**	0.107	MI-202420
41.275(1 5/8)	50.800(2)	12.700(1/2)	18 000	24 900	1 840	2 540	4 000	6 000	**DCL 268**	0.046	MI-222610
	50.800(2)	15.875(5/8)	22 100	32 500	2 260	3 300	4 000	6 000	**DCL2610**	0.058	MI-222610
	50.800(2)	25.400(1)	38 500	66 500	3 950	6 800	4 000	6 000	**DCL2616**	0.106	—
	50.800(2)	31.750(1 1/4)	49 000	90 000	5 000	9 150	4 000	6 000	**DCL2620**	0.116	MI-212620

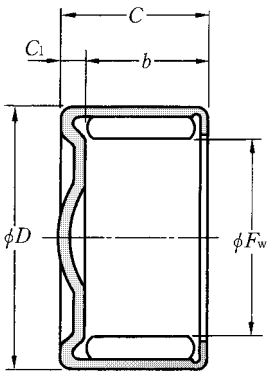
Note 1) Bearing with inner ring is represented by DCL-MI. (Refer to Inner Ring Dimension Table on page B-130.)
EX. DCL2414 + MI-202416
Remarks: Manufacture of the closed end type bearings under this Table is also available.

F_w 44.450～50.800mm

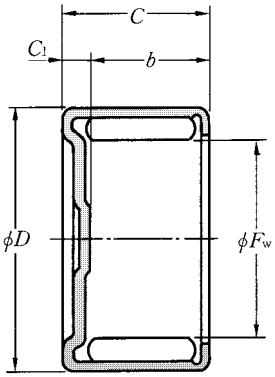
Boundary dimensions mm ($\frac{1}{25.4}$mm)			Basic load ratings N dynamic		Basic load ratings kgf dynamic		Limiting speeds r/min grease		Bearing numbers	Mass kg	Appropriate[1] inner ring
F_w	D	C $^{0}_{-0.2}$	C_r	C_{or}	C_r	C_{or}		oil		(approx.)	(as a reference)
	53.975(2 1/8)	19.050(3/4)	29 200	47 500	2 980	4 850	3 700	5 500	**DCL2812**	0.074	MI-242812
44.450(1 3/4)	53.975(2 1/8)	25.400(1)	40 500	72 000	4 100	7 350	3 700	5 500	**DCL2816**	0.099	MI-242816
	53.975(2 1/8)	38.100(1 1/2)	62 000	126 000	6 350	12 800	3 700	5 500	**DCL2824**	0.149	—
	57.150(2 1/4)	12.700(1/2)	19 700	29 200	2 000	2 980	3 300	5 000	**DCL 308**	0.053	—
47.625(1 7/8)	57.150(2 1/4)	15.875(5/8)	24 200	38 000	2 460	3 900	3 300	5 000	**DCL3010**	0.066	—
	57.150(2 1/4)	25.400(1)	42 000	78 000	4 300	7 950	3 300	5 000	**DCL3016**	0.106	—
	60.325(2 3/8)	12.700(1/2)	20 400	31 500	2 080	3 200	3 100	4 700	**DCL 328**	0.056	—
50.800(2)	60.325(2 3/8)	25.400(1)	44 000	83 500	4 450	8 550	3 100	4 700	**DCL3216**	0.112	—
	60.325(2 3/8)	31.750(1 1/4)	55 500	113 000	5 650	11 500	3 100	4 700	**DCL3220**	0.140	—
	60.325(2 3/8)	38.100(1 1/2)	67 500	146 000	6 850	14 800	3 100	4 700	**DCL3224**	0.168	—

Note 1) Bearing with inner ring is represented by DCL-MI. (Refer to Inner Ring Dimension Table on page B-130.)
EX. DCL2816 + MI-242816
Remarks: Manufacture of the closed end type bearings under this Table is also available.

Type HCK



Standard type



Waffle type

F_w 10～20mm

Boundary dimensions mm					Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Bearing numbers		Mass kg
F_w	D	C	b	C_1	C_r	C_{or}	C_r	C_{or}	standard type	waffle type	(approx.)
10	15	9.35	7.6	1.75	6 200	9 250	635	940	— —	**HCK1015**	0.007
11.656	17.1	11.85	9.6	2.25	8 850	13 800	905	1 400	**HCK1217**	— —	0.011
13	19	11.85	9.6	2.25	10 000	15 000	1 020	1 530	**HCK1319**	— —	0.013
14	20	11.85	9.6	2.25	10 500	16 600	1 070	1 690	**HCK1420**	— —	0.014
16	22	12.85	10.6	2.25	12 200	20 700	1 240	2 110	**HCK1622**	— —	0.017
18	24	13.85	11.6	2.25	13 900	25 300	1 420	2 580	**HCK1824**	— —	0.021
18	24.6	13.85	11.6	2.25	13 900	25 300	1 420	2 580	**HCK1825**	— —	0.025
20	27.9	15.82	13.1	2.72	17 800	31 000	1 810	3 150	**HCK2028**	— —	0.037

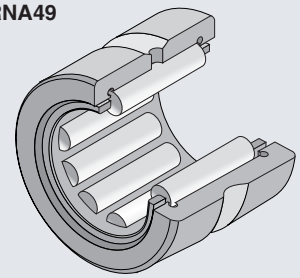
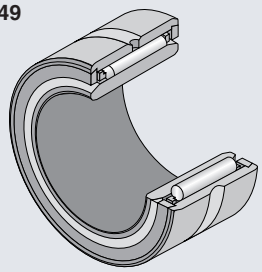
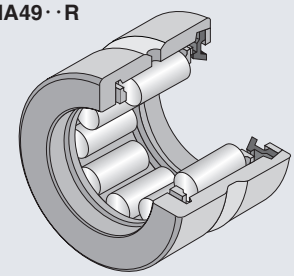
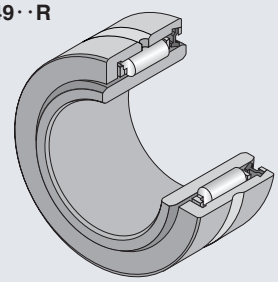
Machined Ring Needle Roller Bearings

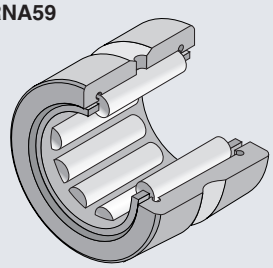
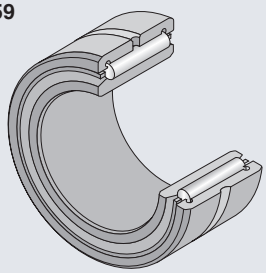
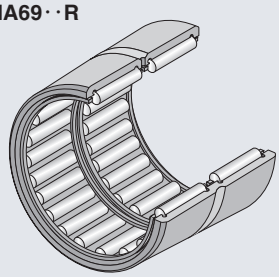
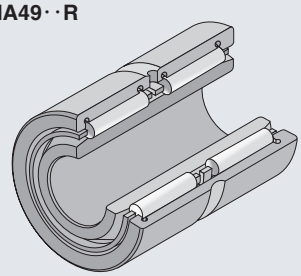


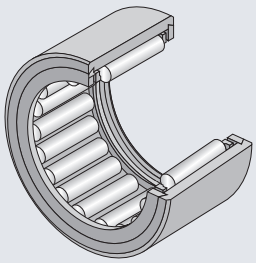
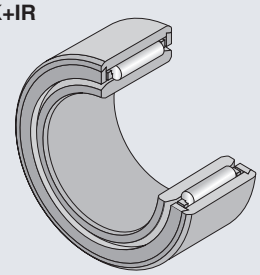
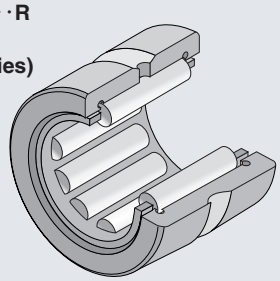
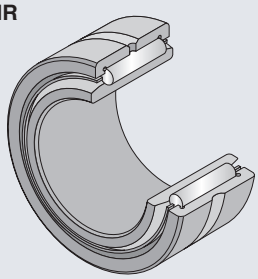
Machined Ring Needle Roller Bearings

The machined ring of this bearing type contains needle rollers and a cage. And the outer ring and the needle rollers are inseparable from each other by means of double-side ribs on the outer ring or side plates.

Because of its machined (solid) outer ring enabling to make it more rigid and to thereby upgrade the bearing accuracy, this bearing type is suitable for an application requiring high speed, high load and high running accuracy. These machined ring needle roller bearings are available in two types -- one without inner ring and another with inner ring -- considering the case of using a shaft as the direct raceway surface without using inner ring.

Bearing type	Applied shaft diameter (mm)	Composition of nominal bearing number	Remarks
Type RNA49 	ϕ7～ϕ12	**RNA 49 5 T2** Tail code Bore dia. No. Dimension series Type code **[Tail code]** T2: Resin cage	Bearing with tail code T2 uses a polyamide resin cage and, therefore, it shall be used at allowable temperature 120˚C and, under continuous running, at 100˚C and less. The dimension series is in compliance with JIS B 1512 or ISO 15.
Type NA49 	ϕ5～ϕ9		
Type RNA49··R 	ϕ14～ϕ490 With seal ϕ14～ϕ58	**RNA 49 02 R** Tail code Bore dia. No. Dimension series Type code **[Tail code]** R: Rib type L: Single-side seal type LL: Double-side seal type	Bearing with seal type (Tail code: L or LL) - synthetic rubber seal built in at single side or double sides is internally filled up with ithium soap base grease. The sealed bearing type shall be used within the temperature range of -25 to 100˚C to prevent deterioration of the seal and filled-up grease. The dimension series is in compliance with JIS B 15 or ISO 15.
Type NA49··R 	ϕ10～ϕ440 With seal ϕ10～ϕ50		

Bearing type	Applied shaft diameter (mm)	Composition of nominal bearing number	Remarks
Type RNA59 	ϕ20～ϕ160	RNA 59 02 Type No. Dimension series Type code	The dimension series is in compliance with JIS B 1512 or ISO 15.
Type NA59 	ϕ15～ϕ140	NA 59 ／22 Type No. Dimension series Type code	
Type RNA69··R 	ϕ16～ϕ35	RNA 69 01 R Tail code Type No. Dimension series Type code	
	ϕ40～ϕ110 Built-in cage double-row type		
Type NA49··R 	ϕ12～ϕ30	NA 69 ／22 Type No. Dimension series Type code	
	ϕ32～ϕ95 Built-in cage double-row type		

Bearing type	Applied shaft diameter (mm)	Composition of nominal bearing number	Remarks
Type NK 	ϕ5～ϕ12	**NK 7 ／ 10 T2** Tail code Width Inscribed circle diameter Type code	Bearing with tail code T2 uses a polyamide resin cage and, therefore, it shall be used at allowable temperature 120˚C and, under continuous running, at 100˚C and less.
Type NK+IR 	ϕ5～ϕ9	**NK24／16R＋ IR 20 × 24 × 16** Type code Bore diameter Outer diameter Width **[Tail code]** R: Rib type T2: Resin cage	
Type NK··R **Type MR** **(Inch series)** 	NK ϕ14～ϕ165	**MR 10 18 12** Width code Outer diameter code Inscribed circle dia. code Type code	
	MR ϕ15.875～ ϕ234.95		
NK··Type R+IR **Type MR+MI** **(Inch series)** 	NK · · R＋IR ϕ10～ϕ150	**MR101812＋ MI－ 06 10 12** Type code Inscribed circle dia. code Outer diameter code Width code	

Bearing Tolerances

The dimensional accuracy, profile accuracy and running accuracy of machined ring needle roller bearings are specified in JIS B 1514 (Accuracy of Rolling Bearings). (Refer to Section 4 "**Bearing Tolerances**", **Table 4.3** on page A-26.) Although the accuracy of **NTN** standard bearings conforms to JIS Class-0, **NTN** can also supply bearings conforming to JIS Class-6, -5 and -4.

Feel free to contact **NTN** for the further detail of these bearings.

The dimensional tolerances for the roller inscribed circle diameters (F_w) of bearing **type NK**, **RNA** and **MR** conform to ISO Tolerance Range Class-F6. Refer to this ISO Tolerance Class

in selecting shaft diameter and radial clearance.

Radial clearance and bearing fits

NTN machined ring needle roller bearings (with inner ring) are all manufactured within the radial clearance range specified in **Table 5.1** of Subsection 5.1 "**Radial Clearance in Bearings**" (page A-30). Because of the narrow non-interchangeable clearance range, the bearings shipped after adjusted to a specific non-interchangeable clearance must be installed with the clearance remained unchanged.

The respective dimensional tolerances for shaft and housing bore sizes on/in which bearing with inner ring is press-fitted are per **Table 6,3** of 6.4 "**Recommended Bearing Fits**" (page A-34) which specifies them according to load characteristic, load magnitude, and shaft and housing bore sizes.The profile accuracy and

surface roughness of shaft and housing to be applied are as specified in **Table 7.3** of 7.3 "**Shaft and Housing Accuracy**" (page A-39).

A shaft is used as the direct raceway surface for a bearing without inner ring and, in this case, the shaft diameter (raceway diameter) tolerance is per **Table 1** specifying the tolerances according to each running clearance. And K7 tolerance generally in broad use is applied as the dimensional tolerance for the housing bore. Feel free to contact **NTN** for application of housing bore tolerance other than K7.

For the profile accuracy, surface roughness and hardness of a shaft acting as the raceway surface, refer to **Table 7.4** of 7.4 "**Raceway Surface Accuracy**"(page A-39).

Table 1 Shaft diameter (raceway diameter) tolerance

Roller inscribed circle dia. F_w mm		Tolerance range class for shaft		
Over	incl.	Smaller than ordinary clearance	Ordinary class	Larger than ordinary clearance
	80	k5	h5	f6
80	160	k5	g5	f6
160	180	k5	g5	e6
180	200	j5	g5	e6
200	250	j5	f6	e6
250	315	h5	f6	e6
315	400	g5	f6	d6

Dimension of oil hole in outer ring

The outer ring is provided with an oil hole and an oil groove to facilitate oil lubrication to bearing. (but excluding Type NK with roller inscribed circle diameter (F_w) of 12mm and less.)

Table 2 shows the oil hole dimension every outer ring diameter.

Table 2 Oil hole dimension

Outer ring outer diameter		Oil hole dia. (mm)	Number of oil hole
Over	incl.		
—	20	2.0	1
20	40	2.5	1
40	80	3.0	1
80	200	3.5	1
200	350	4.0	1
350	—	5.0	1

Mounting relations

In the case of raceway with an oil hole, the bearing must be installed so the oil hole locates within the non-load area. In addition, any bearing with inner ring must be used within the allowable stroking value (s) (with the rollers retained within the effective contact length range of inner ring).

For the allowable stroking value (s), refer to Fig. 1 and applicable "Dimensions Table".

The inner ring and outer ring of machined ring needle roller bearing must be positioned in axial direction by shoulder or snap ring in this case. For the shoulder dimension and corner roundness (r_a) of the shaft and the housing are as specified in applicable "Dimensions Table".

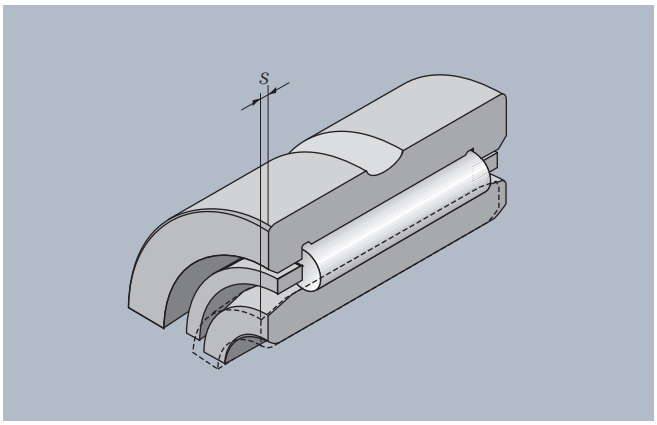
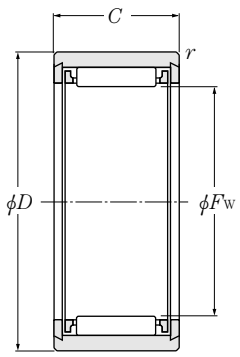


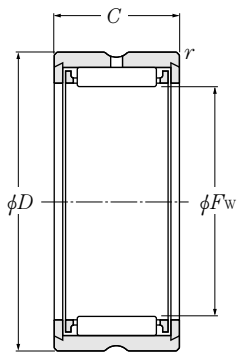
Fig. 1

Without Inner Ring

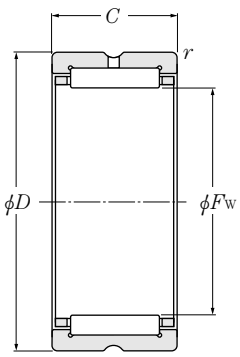
Type RNA49
Type RNA59
Type RNA69
Type NK



Type NK
(F_W≦12mm)



Type RNA49(F_W≦12mm)

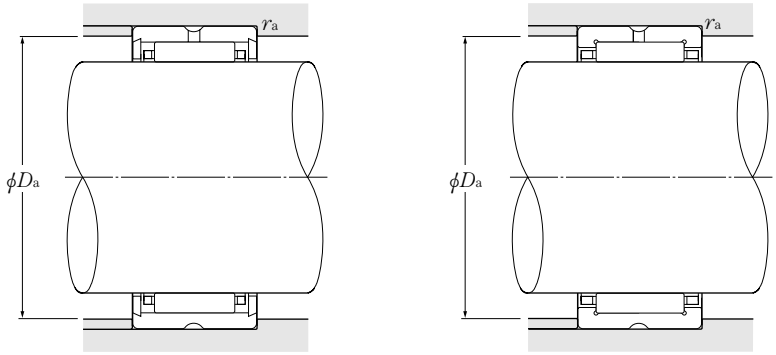


Type RNA49··R(F_W≧14mm)
Type RNA59
Type RNA69··R
Type NK··R(F_W≧14mm)

F_W 5～16mm

Boundary dimensions mm				Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	oil	Bearing numbers	Abutment dimensions mm		Mass kg
F_w	D	C	$r_{s\,min}$ [1]	C_r	C_{or}	C_r	C_{or}				D_a max	r_{as} max	(approx.)
5 +0.018 +0.010	10	10	0.15	2 640	2 190	269	224	27 000	40 000	NK5/10T2	6.5	0.15	0.0031
	10	12	0.15	2 720	2 250	277	230	27 000	40 000	NK5/12T2	6.5	0.15	0.0037
6 +0.018 +0.010	12	10	0.15	2 660	2 280	272	233	25 000	37 000	NK6/10T2	7.5	0.15	0.0047
	12	12	0.15	3 400	3 150	345	320	25 000	37 000	NK6/12T2	7.5	0.15	0.0057
7 +0.022 +0.013	13	10	0.15	2 670	2 350	272	239	23 000	34 000	RNA495T2	8.5	0.15	0.0055
	14	10	0.3	2 670	2 350	272	239	23 000	34 000	NK7/10T2	8.5	0.3	0.0069
	14	12	0.3	3 400	3 200	345	330	23 000	34 000	NK7/12T2	8.5	0.3	0.0082
8 +0.022 +0.013	15	10	0.15	3 150	3 000	320	305	21 000	32 000	RNA496	9.5	0.15	0.0073
	15	12	0.3	4 000	4 100	410	420	21 000	32 000	NK8/12T2	9.5	0.3	0.0087
	15	16	0.3	4 850	5 200	495	535	21 000	32 000	NK8/16T2	9.5	0.3	0.0120
9 +0.022 +0.013	16	12	0.3	4 550	5 000	465	510	20 000	30 000	NK9/12T2	10.5	0.3	0.0100
	16	16	0.3	5 500	6 400	560	650	20 000	30 000	NK9/16T2	10.5	0.3	0.0130
	17	10	0.15	3 600	3 650	365	375	20 000	30 000	RNA497	10.5	0.15	0.0095
10 +0.022 +0.013	17	12	0.3	4 550	5 100	460	520	19 000	28 000	NK10/12T2	11.5	0.3	0.0100
	17	16	0.3	5 450	6 450	555	660	19 000	28 000	NK10/16	11.5	0.3	0.0130
	19	11	0.15	4 300	3 950	435	405	19 000	28 000	RNA498	12	0.15	0.0130
12 +0.027 +0.016	19	12	0.3	5 000	6 100	510	620	17 000	26 000	NK12/12	13.5	0.3	0.0130
	19	16	0.3	6 000	7 700	615	785	17 000	26 000	NK12/16	13.5	0.3	0.0160
	20	11	0.3	4 850	4 900	495	500	17 000	26 000	RNA499	14	0.3	0.0130
14 +0.027 +0.016	22	13	0.3	8 600	9 200	875	935	16 000	24 000	RNA4900R	20	0.3	0.0170
	22	16	0.3	10 300	11 500	1 050	1 170	16 000	24 000	NK14/16R	20	0.3	0.0210
	22	20	0.3	13 000	15 600	1 330	1 590	16 000	24 000	NK14/20R	20	0.3	0.0260
15 +0.027 +0.016	23	16	0.3	10 900	12 700	1 110	1 290	15 000	23 000	NK15/16R	21	0.3	0.0220
	23	20	0.3	13 800	17 200	1 410	1 750	15 000	23 000	NK15/20R	21	0.3	0.0270
16 +0.027 +0.016	24	13	0.3	9 550	10 900	975	1 110	15 000	23 000	RNA4901R	22	0.3	0.0170
	24	16	0.3	12 200	14 900	1 240	1 520	15 000	23 000	NK16/16R	22	0.3	0.0220
	24	20	0.3	14 600	18 800	1 490	1 920	15 000	23 000	NK16/20R	22	0.3	0.0280
	24	22	0.3	15 400	20 000	1 570	2 040	15 000	23 000	RNA6901R	22	0.3	0.0310

Note 1) Allowable minimum chamfer dimension r.



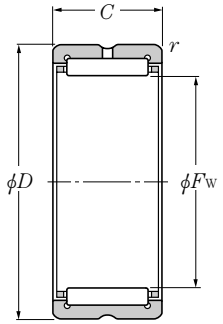
F_w 17～28mm

Boundary dimensions mm				Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	oil	Bearing numbers	Abutment dimensions mm		Mass kg
F_w	D	C	$r_{s\,min}$ [1]	C_r	C_{or}	C_r	C_{or}				D_a max	r_{as} max	(approx.)
17 $^{+0.027}_{+0.016}$	25	16	0.3	12 100	15 000	1 240	1 530	15 000	22 000	**NK17/16R**	23	0.3	0.024
	25	20	0.3	15 400	20 400	1 570	2 080	15 000	22 000	**NK17/20R**	23	0.3	0.030
18 $^{+0.027}_{+0.016}$	26	16	0.3	12 700	16 200	1 300	1 650	14 000	21 000	**NK18/16R**	24	0.3	0.025
	26	20	0.3	16 100	22 000	1 640	2 250	14 000	21 000	**NK18/20R**	24	0.3	0.031
19 $^{+0.033}_{+0.020}$	27	16	0.3	13 300	17 400	1 350	1 780	14 000	21 000	**NK19/16R**	25	0.3	0.026
	27	20	0.3	16 000	22 200	1 630	2 260	14 000	21 000	**NK19/20R**	25	0.3	0.032
20 $^{+0.033}_{+0.020}$	28	13	0.3	10 300	12 800	1 050	1 310	13 000	20 000	**RNA4902R**	26	0.3	0.022
	28	16	0.3	13 200	17 500	1 340	1 790	13 000	20 000	**NK20/16R**	26	0.3	0.027
	28	18	0.3	14 100	19 100	1 440	1 950	13 000	20 000	**RNA5902**	26	0.3	0.033
	28	20	0.3	16 700	23 800	1 700	2 420	13 000	20 000	**NK20/20R**	26	0.3	0.034
	28	23	0.3	17 600	25 300	1 790	2 580	13 000	20 000	**RNA6902R**	26	0.3	0.040
21 $^{+0.033}_{+0.020}$	29	16	0.3	13 700	18 700	1 400	1 910	13 000	19 000	**NK21/16R**	27	0.3	0.028
	29	20	0.3	18 300	27 100	1 860	2 760	13 000	19 000	**NK21/20R**	27	0.3	0.035
22 $^{+0.033}_{+0.020}$	30	16	0.3	14 200	19 900	1 450	2 030	12 000	18 000	**NK22/16R**	28	0.3	0.034
	30	20	0.3	18 000	27 000	1 840	2 760	12 000	18 000	**NK22/20R**	28	0.3	0.037
	30	13	0.3	11 200	14 600	1 140	1 490	12 000	18 000	**RNA4903R**	28	0.3	0.022
	30	18	0.3	15 200	21 700	1 550	2 210	12 000	18 000	**RNA5903**	28	0.3	0.035
	30	23	0.3	18 200	27 200	1 850	2 770	12 000	18 000	**RNA6903R**	28	0.3	0.042
24 $^{+0.033}_{+0.020}$	32	16	0.3	15 200	22 300	1 550	2 280	11 000	170 00	**NK24/16R**	30	0.3	0.032
	32	20	0.3	18 600	28 800	1 890	2 930	11 000	17 000	**NK24/20R**	30	0.3	0.040
25 $^{+0.033}_{+0.020}$	33	16	0.3	15 100	22 400	1 540	2 280	11 000	16 000	**NK25/16R**	31	0.3	0.033
	33	20	0.3	19 200	30 500	1 960	3 100	11 000	16 000	**NK25/20R**	31	0.3	0.042
	37	17	0.3	21 300	25 500	2 170	2 600	11 000	16 000	**RNA4904R**	35	0.3	0.052
	37	23	0.3	28 400	37 000	2 900	3 750	11 000	16 000	**RNA5904**	35	0.3	0.084
	37	30	0.3	36 500	50 500	3 700	5 150	11 000	16 000	**RNA6904R**	35	0.3	0.100
26 $^{+0.033}_{+0.020}$	34	16	0.3	15 600	23 600	1 590	2 410	10 000	15 000	**NK26/16R**	32	0.3	0.034
	34	20	0.3	19 100	30 500	1 940	3 100	10 000	15 000	**NK26/20R**	32	0.3	0.042
28 $^{+0.033}_{+0.020}$	37	20	0.3	22 300	34 000	2 280	3 450	9 500	14 000	**NK28/20R**	35	0.3	0.052
	37	30	0.3	26 700	48 000	2 720	4 900	9 500	14 000	**NK28/30R**	35	0.3	0.082

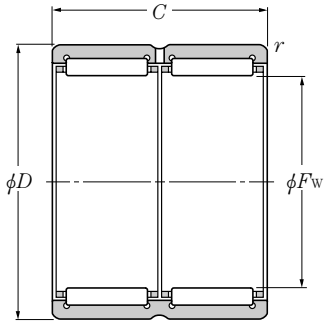
Note 1) Allowable minimum chamfer dimension r.

Without Inner Ring

Type RNA49
Type RNA59
Type RNA69
Type NK



Type RNA49··R
Type RNA59
Type RNA69··R(F_w≦35mm)
Type NK··R

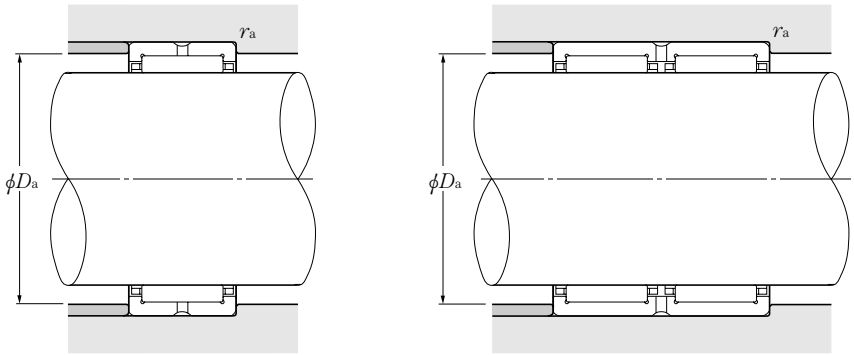


Type RNA69··Type R(F_w≧40mm)

F_w 28～40mm

Boundary dimensions mm				Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	oil	Bearing numbers	Abutment dimensions mm		Mass kg
F_w	D	C	$r_{s\,min}$ [1]	C_r	C_{or}	C_r	C_{or}				D_a max	r_{as} max	(approx.)
	39	17	0.3	23 200	29 300	2 360	2 990	9 500	14 000	RNA49/22R	37	0.3	0.050
28 +0.033 +0.020	39	23	0.3	26 400	37 500	2 690	3 850	9 500	14 000	RNA59/22	37	0.3	0.092
	39	30	0.3	40 000	58 500	4 050	6 000	9 500	14 000	RNA69/22R	37	0.3	0.100
29 +0.033 +0.020	38	20	0.3	22 200	34 000	2 270	3 450	9 500	14 000	NK29/20R	36	0.3	0.054
	38	30	0.3	27 500	50 500	2 810	5 150	9 500	14 000	NK29/30R	36	0.3	0.084
	40	20	0.3	22 100	34 000	2 260	3 500	8 500	13 000	NK30/20R	38	0.3	0.065
	40	30	0.3	33 000	57 000	3 350	5 800	8 500	13 000	NK30/30R	38	0.3	0.098
30 +0.033 +0.020	42	17	0.3	24 000	31 500	2 450	3 200	8 500	13 000	RNA4905R	40	0.3	0.061
	42	23	0.3	30 500	43 000	3 150	4 350	8 500	13 000	RNA5905	40	0.3	0.101
	42	30	0.3	41 500	63 000	4 200	6 400	8 500	13 000	RNA6905R	40	0.3	0.112
	42	20	0.3	23 500	37 500	2 400	3 850	8 500	13 000	NK32/20R	40	0.3	0.068
	42	30	0.3	34 000	60 500	3 450	6 150	8 500	13 000	NK32/30R	40	0.3	0.102
32 +0.041 +0.025	45	17	0.3	24 800	33 500	2 530	3 400	8 500	13 000	RNA49/28R	43	0.3	0.073
	45	23	0.3	32 000	45 500	3 250	4 650	8 500	13 000	RNA59/28	43	0.3	0.108
	45	30	0.3	43 000	67 000	4 350	6 850	8 500	13 000	RNA69/28R	43	0.3	0.135
	45	20	0.3	24 800	41 500	2 520	4 250	7 500	11 000	NK35/20R	43	0.3	0.074
	45	30	0.3	36 000	66 500	3 650	6 800	7 500	11 000	NK35/30R	43	0.3	0.112
35 +0.041 +0.025	47	17	0.3	25 500	35 500	2 600	3 600	7 500	11 000	RNA4906R	45	0.3	0.069
	47	23	0.3	32 500	48 500	3 350	4 950	7 500	11 000	RNA5906	45	0.3	0.108
	47	30	0.3	42 500	67 500	4 300	6 900	7 500	11 000	RNA6906R	45	0.3	0.126
37 +0.041 +0.025	47	20	0.3	25 300	43 500	2 580	4 400	7 500	11 000	NK37/20R	45	0.3	0.077
	47	30	0.3	36 500	69 500	3 750	7 100	7 500	11 000	NK37/30R	45	0.3	0.107
38 +0.041 +0.025	48	20	0.3	25 900	45 000	2 640	4 600	7 500	11 000	NK38/20R	46	0.3	0.079
	48	30	0.3	37 500	73 000	3 850	7 400	7 500	11 000	NK38/30R	46	0.3	0.107
	50	20	0.3	26 400	47 000	2 700	4 800	6 500	10 000	NK40/20R	48	0.3	0.083
	50	30	0.3	38 500	76 000	3 900	7 750	6 500	10 000	NK40/30R	48	0.3	0.125
40 +0.041 +0.025	52	20	0.6	31 500	47 500	3 200	4 850	6 500	10 000	RNA49/32R	48	0.6	0.089
	52	27	0.6	38 000	61 000	3 850	6 250	6 500	10 000	RNA59/32	48	0.6	0.149
	52	36	0.6	47 500	82 000	4 850	8 350	6 500	10 000	RNA69/32R	48	0.6	0.162

Note 1) Allowable minimum chamfer dimension r.



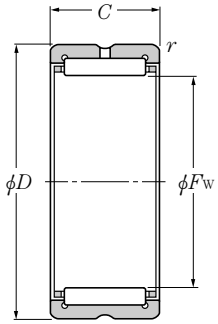
F_W 42～63mm

Boundary dimensions mm				Basic load ratings N dynamic		kgf dynamic		Limiting speeds r/min grease		Bearing numbers	Abutment dimensions mm		Mass kg
F_W	D	C	$r_{s\,min}$ [1]	C_r	C_{or}	C_r	C_{or}		oil		D_a max	r_{as} max	(approx.)
	52	20	0.3	26 900	49 000	2 750	5 000	6 500	9 500	**NK42/20R**	50	0.3	0.086
	52	30	0.3	39 000	79 000	4 000	8 050	6 500	9 500	**NK42/30R**	50	0.3	0.130
42 $^{+0.041}_{+0.025}$	55	20	0.6	32 000	50 000	3 300	5 100	6 500	9 500	**RNA4907R**	51	0.6	0.107
	55	27	0.6	39 000	64 500	3 950	6 550	6 500	9 500	**RNA5907**	51	0.6	0.176
	55	36	0.6	49 000	86 500	5 000	8 800	6 500	9 500	**RNA6907R**	51	0.6	0.193
43 $^{+0.041}_{+0.025}$	53	20	0.3	27 500	51 000	2 810	5 200	6 500	9 500	**NK43/20R**	51	0.3	0.086
	53	30	0.3	40 000	82 000	4 100	8 400	6 500	9 500	**NK43/30R**	51	0.3	0.133
45 $^{+0.041}_{+0.025}$	55	20	0.3	28 000	52 500	2 860	5 400	6 000	9 000	**NK45/20R**	53	0.3	0.092
	55	30	0.3	41 000	85 500	4 150	8 700	6 000	9 000	**NK45/30R**	53	0.3	0.139
47 $^{+0.041}_{+0.025}$	57	20	0.3	28 800	55 500	2 940	5 650	5 500	8 500	**NK47/20R**	55	0.3	0.095
	57	30	0.3	42 500	91 500	4 350	9 350	5 500	8 500	**NK47/30R**	55	0.3	0.142
	62	22	0.6	43 500	66 500	4 450	6 800	5 500	8 500	**RNA4908R**	58	0.6	0.140
48 $^{+0.041}_{+0.025}$	62	30	0.6	53 000	92 500	5 450	9 450	5 500	8 500	**RNA5908**	58	0.6	0.225
	62	40	0.6	67 000	116 000	6 850	11 800	5 500	8 500	**RNA6908R**	58	0.6	0.256
50 $^{+0.041}_{+0.025}$	62	25	0.6	38 500	74 500	3 950	7 550	5 500	8 000	**NK50/25R**	58	0.6	0.158
	62	35	0.6	51 000	106 000	5 200	10 800	5 500	8 000	**NK50/35R**	58	0.6	0.221
	68	22	0.6	46 000	73 000	4 700	7 450	5 000	7 500	**RNA4909R**	64	0.6	0.182
52 $^{+0.049}_{+0.030}$	68	30	0.6	56 000	101 000	5 700	10 300	5 000	7 500	**RNA5909**	64	0.6	0.232
	68	40	0.6	70 500	127 000	7 200	13 000	5 000	7 500	**RNA6909R**	64	0.6	0.273
55 $^{+0.049}_{+0.030}$	68	25	0.6	41 000	82 000	4 150	8 400	5 000	7 500	**NK55/25R**	64	0.6	0.193
	68	35	0.6	54 000	118 000	5 500	12 000	5 000	7 500	**NK55/35R**	64	0.6	0.267
	72	22	0.6	48 000	80 000	4 900	8 150	4 700	7 000	**RNA4910R**	68	0.6	0.163
58 $^{+0.049}_{+0.030}$	72	30	0.6	58 000	110 000	5 950	11 200	4 700	7 000	**RNA5910**	68	0.6	0.289
	72	40	0.6	74 000	139 000	7 500	14 200	4 700	7 000	**RNA6910R**	68	0.6	0.320
60 $^{+0.049}_{+0.030}$	72	25	0.6	41 000	85 000	4 200	8 700	4 300	6 500	**NK60/25R**	68	0.6	0.185
	72	35	0.6	57 000	130 000	5 800	13 200	4 300	6 500	**NK60/35R**	68	0.6	0.258
63 $^{+0.049}_{+0.030}$	80	25	1	58 500	99 500	6 000	10 100	4 300	6 500	**RNA4911R**	75	1	0.255
	80	34	1	76 500	140 000	7 800	14 300	4 300	6 500	**RNA5911**	75	1	0.367

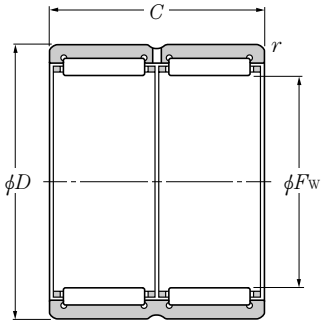
Note 1) Allowable minimum chamfer dimension r.

Without Inner Ring

Type RNA49
Type RNA59
Type RNA69
Type NK



Type RNA48
Type RNA49··R, Type RNA49
RNA59 type
Type NK··R, Type NK

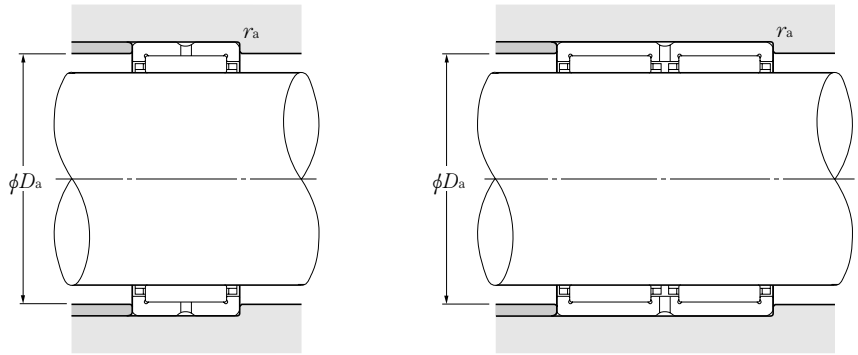


Type RNA69··R

F_w 63～85mm

Boundary dimensions mm				Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	oil	Bearing numbers	Abutment dimensions mm		Mass kg
F_w	D	C	$r_{s\,min}$ [1]	C_r	C_{or}	C_r	C_{or}				D_a max	r_{as} max	(approx.)
63 $^{+0.049}_{+0.030}$	80	45	1	94 000	183 000	9 600	18 600	4 300	6 500	RNA6911R	75	1	0.470
65 $^{+0.049}_{+0.030}$	78	25	0.6	45 000	98 000	4 550	10 000	4 000	6 000	NK65/25R	74	0.6	0.221
	78	35	0.6	60 000	142 000	6 100	14 400	4 000	6 000	NK65/35R	74	0.6	0.310
68 $^{+0.049}_{+0.030}$	82	25	1	44 500	89 000	4 500	9 050	4 000	6 000	NK68/25R	77	0.6	0.241
	82	35	0.6	63 000	139 000	6 400	14 200	4 000	6 000	NK68/35R	78	0.6	0.338
	85	25	1	61 500	108 000	6 250	11 000	4 000	6 000	RNA4912R	80	1	0.275
	85	34	1	80 500	153 000	8 200	15 600	4 000	6 000	RNA5912	80	1	0.408
	85	45	1	95 500	191 000	9 750	19 400	4 000	6 000	RNA6912R	80	1	0.488
70 $^{+0.049}_{+0.030}$	85	25	0.6	45 000	91 500	4 600	9 350	3 700	5 500	NK70/25R	81	0.6	0.275
	85	35	0.6	64 000	144 000	6 550	14 700	3 700	5 500	NK70/35R	81	0.6	0.386
72 $^{+0.049}_{+0.030}$	90	25	1	62 500	112 000	6 350	11 400	3 700	5 500	RNA4913R	85	1	0.312
	90	34	1	84 000	165 000	8 600	16 800	3 700	5 500	RNA5913	85	1	0.462
	90	45	1	97 000	198 000	9 900	20 200	3 700	5 500	RNA6913R	85	1	0.520
73 $^{+0.049}_{+0.030}$	90	25	0.6	54 000	100 000	5 500	10 200	3 700	5 500	NK73/25R	86	0.6	0.302
	90	35	0.6	76 500	156 000	7 800	16 000	3 700	5 500	NK73/35R	86	0.6	0.428
75 $^{+0.049}_{+0.030}$	92	25	0.6	55 000	104 000	5 600	10 600	3 700	5 500	NK75/25R	88	0.6	0.315
	92	35	0.6	78 000	162 000	7 950	16 500	3 700	5 500	NK75/35R	88	0.6	0.492
80 $^{+0.049}_{+0.030}$	95	25	1	57 000	119 000	5 800	12 200	3 300	5 000	NK80/25R	90	1	0.301
	95	35	1	79 500	184 000	8 150	18 700	3 300	5 000	NK80/35R	90	1	0.425
	100	30	1	85 500	156 000	8 750	15 900	3 300	5 000	RNA4914R	95	1	0.460
	100	40	1	103 000	187 000	10 500	19 100	3 300	5 000	RNA5914	95	1	0.706
	100	54	1	130 000	267 000	13 300	27 200	3 300	5 000	RNA6914R	95	1	0.857
85 $^{+0.058}_{+0.036}$	105	25	1	70 500	123 000	7 200	12 600	3 100	4 700	NK85/25R	100	1	0.404
	105	30	1	87 000	162 000	8 900	16 500	3 100	4 700	RNA4915R	100	1	0.489
	105	35	1	100 000	193 000	10 200	19 700	3 100	4 700	NK85/35R	100	1	0.517
	105	40	1	109 000	205 000	11 100	20 900	3 100	4 700	RNA5915	100	1	0.745
	105	54	1	132 000	277 000	13 500	28 300	3 100	4 700	RNA6915R	100	1	0.935

Note 1) Allowable minimum chamfer dimension r.



F_w 90～130mm

Boundary dimensions mm				Basic load ratings N dynamic	static	kgf dynamic	static	Limiting speeds r/min grease	oil	Bearing numbers	Abutment and fillet dimensions mm		Mass kg
F_w	D	B_e	$r_{s\,min}$ [1]	C_r	C_{or}	C_r	C_{or}				D_a max	r_{as} max	(approx.)
	110	25	1	71 500	128 000	7 300	13 100	2 900	4 400	**NK90/25R**	105	1	0.426
	110	30	1	90 500	174 000	9 250	17 700	2 900	4 400	**RNA4916R**	105	1	0.516
90 $^{+0.058}_{+0.036}$	110	35	1	104 000	208 000	10 600	21 200	2 900	4 400	**NK90/35R**	105	1	0.604
	110	40	1	115 000	223 000	11 700	22 700	2 900	4 400	**RNA5916**	105	1	0.787
	110	54	1	137 000	298 000	14 000	30 500	2 900	4 400	**RNA6916R**	105	1	0.987
95 $^{+0.058}_{+0.036}$	115	26	1	74 500	137 000	7 600	14 000	2 800	4 200	**NK95/26R**	110	1	0.364
	115	36	1	108 000	223 000	11 100	22 700	2 800	4 200	**NK95/36R**	110	1	0.652
	120	26	1	73 500	137 000	7 500	14 000	2 700	4 000	**NK100/26R**	115	1	0.487
	120	35	1.1	112 000	237 000	11 500	24 200	2 700	4 000	**RNA4917R**	113.5	1	0.657
100 $^{+0.058}_{+0.036}$	120	36	1	107 000	223 000	11 000	22 800	2 700	4 000	**NK100/36R**	115	1	0.679
	120	46	1.1	137 000	290 000	14 000	29 600	2 700	4 000	**RNA5917**	113.5	1	1.00
	120	63	1.1	169 000	400 000	17 300	41 000	2 700	4 000	**RNA6917R**	113.5	1	1.20
	125	26	1	76 500	147 000	7 800	14 900	2 500	3 800	**NK105/26R**	120	1	0.506
	125	35	1.1	116 000	252 000	11 900	25 700	2 500	3 800	**RNA4918R**	118.5	1	0.697
105 $^{+0.058}_{+0.036}$	125	36	1	111 000	238 000	11 400	24 300	2 500	3 800	**NK105/36R**	120	1	0.713
	125	46	1.1	143 000	310 000	14 600	32 000	2 500	3 800	**RNA5918**	118.5	1	1.04
	125	63	1.1	175 000	425 000	17 900	43 500	2 500	3 800	**RNA6918R**	118.5	1	1.33
	130	30	1.1	97 500	204 000	9 950	20 800	2 400	3 600	**NK110/30R**	123.5	1	0.612
	130	35	1.1	118 000	260 000	12 000	26 500	2 400	3 600	**RNA4919R**	123.5	1	0.719
110 $^{+0.058}_{+0.036}$	130	40	1.1	129 000	292 000	13 100	29 700	2 400	3 600	**NK110/40R**	123.5	1	0.830
	130	46	1.1	149 000	335 000	15 200	34 000	2 400	3 600	**RNA5919**	123.5	1	1.13
	130	63	1.1	177 000	440 000	18 100	45 000	2 400	3 600	**RNA6919R**	123.5	1	1.46
115 $^{+0.058}_{+0.036}$	140	40	1.1	127 000	260 000	12 900	26 500	2 300	3 500	**RNA4920**	133.5	1	1.15
	140	54	1.1	182 000	395 000	18 600	40 500	2 300	3 500	**RNA5920**	133.5	1	1.76
120 $^{+0.058}_{+0.036}$	140	30	1	93 500	210 000	9 550	21 400	2 200	3 300	**RNA4822**	135	1	0.670
	140	40	1.1	113 000	268 000	11 500	27 300	2 200	3 300	**NK120/40**	133.5	1	0.910
125 $^{+0.068}_{+0.043}$	150	40	1.1	131 000	279 000	13 300	28 400	2 100	3 200	**RNA4922**	143.5	1	1.24
	150	54	1.1	193 000	440 000	19 700	45 000	2 100	3 200	**RNA5922**	143.5	1	1.89
130 $^{+0.068}_{+0.043}$	150	30	1	99 500	233 000	10 100	23 800	2 100	3 100	**RNA4824**	145	1	0.730

Note 1) Allowable minimum chamfer dimension r.